

IMO(22nd) – 1981

First Day

1. P is a point inside a given triangle ABC. D, E and F are the feet of the perpendicular from P to the lines BC, CA and AB respectively. Find all P for which $\frac{BC}{PD} + \frac{CA}{PE} + \frac{AB}{PF}$ is least.
2. Let $1 \leq r \leq n$ and consider all subsets of r elements of the set $\{1, 2, \dots, n\}$. Each of these subsets has a smallest member. Let $F(n, r)$ denote the arithmetic mean of these smallest numbers, prove that $F(n, r) = \frac{n+1}{r+1}$.
3. Determine the maximum value of $m^3 + n^3$, where m, n are integers satisfying $m, n \in \{1, 2, \dots, 1981\}$ and $(n^2 - mn - m^2)^2 = 1$.

Second Day

4.
 - (a) For which values of $n > 2$ is there a set of n consecutive positive integers such that the largest number in the set is a divisor of the least common multiple of the remaining $n - 1$ numbers?
 - (b) For which values of $n > 2$ is there exactly one set having the stated property?
5. Three congruent circles have a common point O and lie inside a given triangle. Each circle touches a pair of sides of the triangle. Prove that the incenter and the circumcenter of the triangle and the point O are collinear.
6. The function $f(x, y)$ satisfies the following for all non-negative integers x, y .
 - (a) $f(0, y) = y + 1$.
 - (b) $f(x + 1, 0) = f(x, 1)$.
 - (c) $f(x + 1, y + 1) = f(x, f(x + 1, y))$
 Determine $f(4, 1981)$.

